PRISM Neural GLM-HMM
Poisson GLM-HMM for Simulating Non-Stationary Neural Dynamics

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1 What This Model Is Trying To Do

The script learns two things at the same time from spike-count data:

1. A single connectivity matrix A (who influences whom).
2. A hidden state sequence $z_t \in \{1, \ldots, K\}$ (here $K = 2$), which captures non-stationary modes over time.

It does this with a Poisson GLM-HMM and a two-stage training process:

1. Stage A: find sparse structure (which edges likely exist).
2. Stage B: debias strengths on the frozen structure (reduce over-shrinking).

2 Data and Preprocessing

2.1 Observed arrays

$$Y \in \mathbb{R}_{\geq 0}^{T \times N}, \qquad N = 100, \qquad K = 2. \tag{1}$$

Meaning: $Y_{t,n}$ is spike count of neuron n at time bin t.

2.2 Lagged design matrix

$$X_t = \begin{cases} \mathbf{0}, & t = 1, \\ Y_{t-1}, & t \geq 2. \end{cases} \tag{2}$$

Meaning: the model predicts current spikes from the previous bin's spikes.

2.3 Time bin size

$$\Delta t = 0.1. \tag{3}$$

Meaning: rates are converted to expected counts per bin using Δt.

3 Model Parameters

$$A \in \mathbb{R}^{N\times N}, \quad B \in \mathbb{R}^{K\times N}, \quad \pi \in \Delta^{K-1}, \quad P \in \mathbb{R}^{K\times K}. \tag{4}$$

Meaning:

- A: shared connectivity (same across states).
- B_k: state-specific bias vector (different baseline per state).
- π: initial state probabilities.
- P: state transition probabilities.

Orientation convention. A is stored as (post, pre). With row-vector input X_t:

$$\eta_{t,k} = X_t A^\top + B_k. \tag{5}$$

Meaning: each neuron's drive is weighted sum of previous spikes plus state bias.

4 Poisson Emission Model

4.1 Clipped linear predictor

$$\tilde{\eta}_{t,k,n} = \min(\eta_{t,k,n}, 5). \tag{6}$$

Why: prevents numerical explosion from very large positive drives.

4.2 Expected count per bin

$$\mu_{t,k,n} = \exp(\tilde{\eta}_{t,k,n})\,\Delta t. \tag{7}$$

Meaning: Poisson mean count in that bin.

4.3 Emission log score used by E-step

$$\ell_{t,k} = \sum_{n=1}^{N} \left[Y_{t,n}(\tilde{\eta}_{t,k,n} + \log \Delta t) - \mu_{t,k,n} \right]. \tag{8}$$

Meaning: how well state k explains data at time t (ignoring constant $\log(Y_{t,n}!)$ terms).

5 Warm Start Initialization

5.1 Connectivity warm start (ridge)

$$A_{\text{ls}} = (X^\top X + \lambda_{\text{ridge}} I)^{-1} X^\top Y. \tag{9}$$

Meaning: a quick linear estimate before EM.

Only strongest off-diagonals are kept:

$$A_{\text{sparse},ij} = A_{\text{ls},ij}\, \mathbf{1}(|A_{\text{ls},ij}| \geq q_{1-r}(|A_{\text{ls,off}}|))\,, \; i \neq j. \tag{10}$$

Then:

$$A_{\text{sparse},ii} = -0.05, \qquad A_{\text{sparse}} \leftarrow \text{clip}(A_{\text{sparse}}, -0.5, 0.5), \qquad A \leftarrow A_{\text{sparse}}^\top. \tag{11}$$

Meaning: start sparse, stable, and in correct orientation.

5.2 Bias warm start

$$B_{k,n} = \log\left(\max\left(\frac{1}{T}\sum_{t=1}^{T} Y_{t,n}, 10^{-4}\right)\right) + \epsilon_{k,n}, \quad \epsilon_{k,n} \sim \mathcal{N}(0, 0.05^2). \tag{12}$$

Meaning: initial bias is close to mean firing rate, with small state-breaking noise.

6 E-step (Forward-Backward)

6.1 Forward recursion

$$\log \alpha_{1,k} = \ell_{1,k} + \log \pi_k, \tag{13}$$

$$\log \alpha_{t,k} = \ell_{t,k} + \log \sum_{j=1}^{K} \exp\left(\log \alpha_{t-1,j} + \log P_{jk}\right), \ t \geq 2. \tag{14}$$

Meaning: probability of data up to t ending in state k.

6.2 Backward recursion

$$\log \beta_{T,k} = 0, \tag{15}$$

$$\log \beta_{t,j} = \log \sum_{k=1}^{K} \exp\left(\log P_{jk} + \ell_{t+1,k} + \log \beta_{t+1,k}\right), \ t \leq T-1. \tag{16}$$

Meaning: probability of future data given current state.

6.3 Posterior responsibilities

$$\log \gamma_{t,k} = \log \alpha_{t,k} + \log \beta_{t,k} - \log \sum_{j=1}^{K} \exp(\log \alpha_{t,j} + \log \beta_{t,j}). \tag{17}$$

$$\gamma_{t,k} = \exp(\log \gamma_{t,k}). \tag{18}$$

Meaning: probability state k is active at time t.

6.4 Temperature smoothing

$$\log \gamma_{t,k} \leftarrow \frac{\log \gamma_{t,k}}{\tau}, \qquad \tau \in [\tau_{\text{start}}, \tau_{\text{end}}]. \tag{19}$$

Meaning: early training uses softer state posteriors (prevents collapse), later becomes sharper.

6.5 Responsibility floor

$$\gamma_{t,k} \leftarrow (1 - Kf)\gamma_{t,k} + f, \qquad f = \min(\gamma_{\text{floor}}, 0.49/K), \qquad \sum_k \gamma_{t,k} = 1. \tag{20}$$

Meaning: each state keeps a minimum mass so one state does not die too early.

6.6 Pairwise posterior for transitions

$$\log \xi_t(i,j) = \log \alpha_{t,i} + \log P_{ij} + \ell_{t+1,j} + \log \beta_{t+1,j} - \log \sum_{a,b} \exp(\cdots), \tag{21}$$

$$\Xi_{ij} = \sum_{t=1}^{T-1} \xi_t(i,j). \tag{22}$$

Meaning: expected transition counts from state i to j.

7 M-step Stage A (Sparse Structure Learning)

7.1 Loss being optimized

For minibatch $\mathcal{B}$:

$$\mathcal{L}_{\text{pois}} = \sum_{t\in\mathcal{B}} \sum_{k=1}^{K} \gamma_{t,k} \sum_{n=1}^{N} \left[\mu_{t,k,n} - Y_{t,n}(\tilde{\eta}_{t,k,n} + \log \Delta t)\right]. \tag{23}$$

$$\mathcal{R}_{L2} = \frac{\lambda_2}{2} \|A\|_F^2. \tag{24}$$

$$\min_{A,B} \ \mathcal{L}_{\text{pois}} + \mathcal{R}_{L2}. \tag{25}$$

Meaning: fit data while keeping weights from growing too large.

7.2 Excitatory / inhibitory row blocks

$$n_{\text{exc}} = \lfloor 0.8N \rfloor. \tag{26}$$

Rows 1:n_{exc} are treated as excitatory block, remaining rows inhibitory block.

7.3 Annealed excitatory proximal threshold

Global epoch fraction:

$$s = \frac{g}{G-1}. \tag{27}$$

Exc proximal scale schedule:

$$\alpha_{\text{exc}}(s) = \alpha_{\text{exc},0} + \big(\alpha_{\text{exc},1} - \alpha_{\text{exc},0}\big) s^{p_{\text{exc}}}. \tag{28}$$

Exc L1 multiplier schedule:

$$m_{\text{exc}}(s) = m_0 + (m_1 - m_0) s^{p_m}, \qquad \lambda_{\text{exc}}^{\text{eff}}(s) = \lambda_{\text{exc}} m_{\text{exc}}(s). \tag{29}$$

Thresholds:

$$\tau_{\text{exc}} = \lambda_{\text{exc}}^{\text{eff}}(s)\, \alpha_{\text{exc}}(s), \qquad \tau_{\text{inh}} = \lambda_{\text{inh}}\, \alpha_{\text{inh}}. \tag{30}$$

Off-diagonal soft-thresholding:

$$A_{ij} \leftarrow \text{sign}(A_{ij}) \max\left(|A_{ij}| - \tau_{\text{block}(i)}, 0\right), \quad i \neq j. \tag{31}$$

Meaning: weak edges are shrunk toward zero; excitatory shrinkage changes over training.

7.4 Dale sign projection

$$A_{ij} \leftarrow \max(A_{ij}, 0), \quad i \leq n_{\text{exc}},\ i \neq j, \tag{32}$$

$$A_{ij} \leftarrow \min(A_{ij}, 0), \quad i > n_{\text{exc}},\ i \neq j. \tag{33}$$

Meaning: excitatory rows stay nonnegative off-diagonal; inhibitory rows stay nonpositive off-diagonal.

7.5 Diagonal inhibitory constraint

$$A_{ii} \leftarrow -\max(|A_{ii}|, d_{\min}). \tag{34}$$

Meaning: self-coupling is always inhibitory with minimum magnitude.

7.6 Spectral stability projection

$$\rho(A) = \max_{\lambda \in \text{eig}(A)} |\lambda|. \tag{35}$$

$$A \leftarrow A \cdot \frac{0.919}{\rho(A)} \quad \text{if } \rho(A) \geq 0.92. \tag{36}$$

Meaning: keeps recurrent dynamics stable.

7.7 Delayed periodic adaptive hard-zero

Apply only when

$$g \geq g_{\text{start}}, \qquad (g - g_{\text{start}}) \bmod q_{\text{hz}} = 0. \tag{37}$$

Adaptive excit floor:

$$\theta_{\text{exc}} = \max\left(\text{Quantile}\left(|A_{\text{exc,off}}|_{>10^{-12}}, q_{\text{exc}}\right), \theta_{\text{exc,min}}\right). \tag{38}$$

Fixed inhibitory floor θ_{inh}. Hard-zero rule:

$$A_{ij} \leftarrow 0 \quad \text{if } i \neq j \text{ and } |A_{ij}| < \theta_{\text{block}(i)}. \tag{39}$$

Then Dale and diagonal constraints are re-applied.

8 State-Dynamics Parameter Updates

8.1 Initial state distribution

$$\pi_k \leftarrow \frac{\gamma_{1,k} + \alpha_\pi}{\sum_j (\gamma_{1,j} + \alpha_\pi)}. \tag{40}$$

Meaning: smoothed estimate of start-state probabilities.

8.2 Transition matrix

$$\tilde{P}_{ij} = \Xi_{ij} + \epsilon_P + \kappa \delta_{ij}, \qquad P_{ij} = \frac{\tilde{P}_{ij}}{\sum_b \tilde{P}_{ib}}. \tag{41}$$

Meaning: observed transitions + smoothing + sticky self-transition.

9 M-step Stage B (Debias Refit)

After Stage A, freeze off-diagonal support:

$$S_{ij} = \mathbf{1}(|A_{ij}| > \varepsilon_{\text{sup}}), \quad i \neq j. \tag{42}$$

Stage B objective (no L1/prox/hard-zero):

$$\min_{A,B} \ \mathcal{L}_{\text{pois}}(\gamma^{\text{db}}) + \frac{\lambda_2}{2}\|A\|_F^2. \tag{43}$$

After each Stage B gradient step:

$$A_{\text{off}} \leftarrow A_{\text{off}} \odot S, \tag{44}$$

then re-apply Dale projection, spectral projection, and inhibitory diagonal rule.

Responsibilities are refreshed every r_{db} epochs:

$$\gamma^{\text{db}} \leftarrow \text{E-step}(A, B, P, \pi). \tag{45}$$

Early stopping metric in Stage B:

$$F1_{\text{db}}(e) = \frac{2TP(e)}{2TP(e) + FP(e) + FN(e)}. \tag{46}$$

Improve condition:

$$F1_{\text{db}}(e) > F1_{\text{best}} + \delta. \tag{47}$$

If true: save best state and reset patience. Else increment patience. Stop when patience hits p_{db}, then restore best saved (A, B).

10 Final Evaluation and Selection

10.1 Adaptive threshold search per block

$$\mathcal{T}_{\text{block}} = \left\{ \text{Quantile}\left(|\hat{A}_{\text{block}}|_{>10^{-12}}, q\right) : q \in [0.70, 0.995] \right\}. \tag{48}$$

$$\tau_{\text{block}} = \arg\max_{\tau \in \mathcal{T}_{\text{block}}} \frac{2TP_\tau}{2TP_\tau + FP_\tau + FN_\tau}. \tag{49}$$

Meaning: threshold is chosen from model's own nonzero scale, not fixed globally.

10.2 Global F1

$$F1 = \frac{2TP}{2TP + FP + FN}. \tag{50}$$

10.3 Occupancy gate

$$\bar{\gamma}_k = \frac{1}{T}\sum_{t=1}^{T} \gamma_{t,k}, \qquad \text{min_state_occupancy} = \min_k \bar{\gamma}_k. \tag{51}$$

Accept config only if

$$\min_k \bar{\gamma}_k \geq 0.10. \tag{52}$$

11 Results and Visualizations

11.1 Ground Truth Specification

The synthetic dataset uses a 100-neuron recurrent network with 80 excitatory neurons and 20 inhibitory neurons. Rows are post-synaptic neurons and columns are pre-synaptic neurons. Rows 0–79 correspond to excitatory neurons and rows 80–99 correspond to inhibitory neurons.

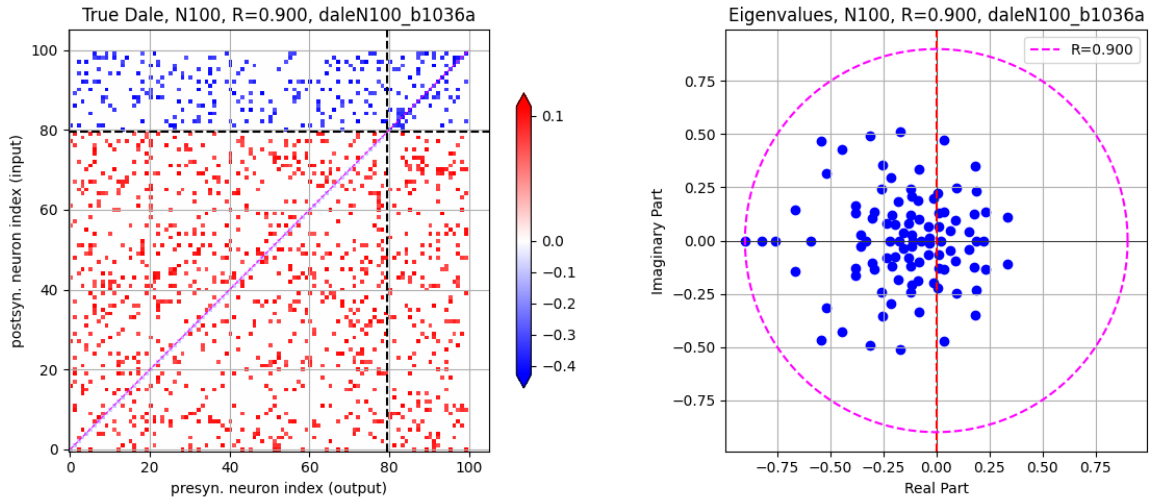

Figure 1: Ground-truth connectivity and spectral stability. Left: true A matrix with Dale-consistent structure. Right: eigenvalue cloud with spectral radius near 0.90, confirming stable simulated dynamics.

State-specific baseline excitability is encoded by the bias vectors B_0 and B_1.

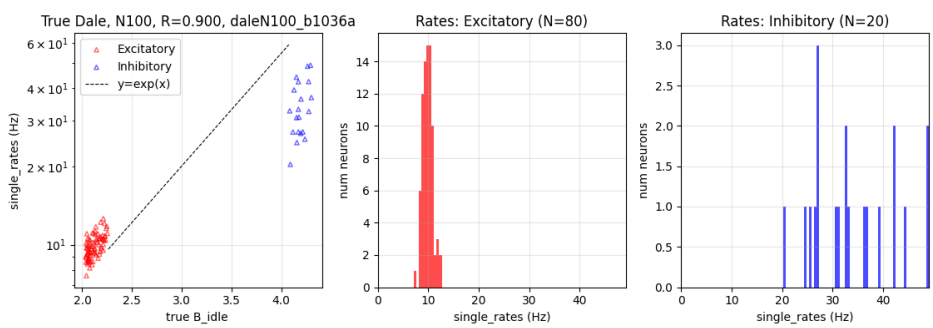

Figure 2: Ground-truth baseline terms. The two state-specific bias vectors create the two latent regimes used by the simulator.

The full synthetic spike sequence and true state schedule are shown below.

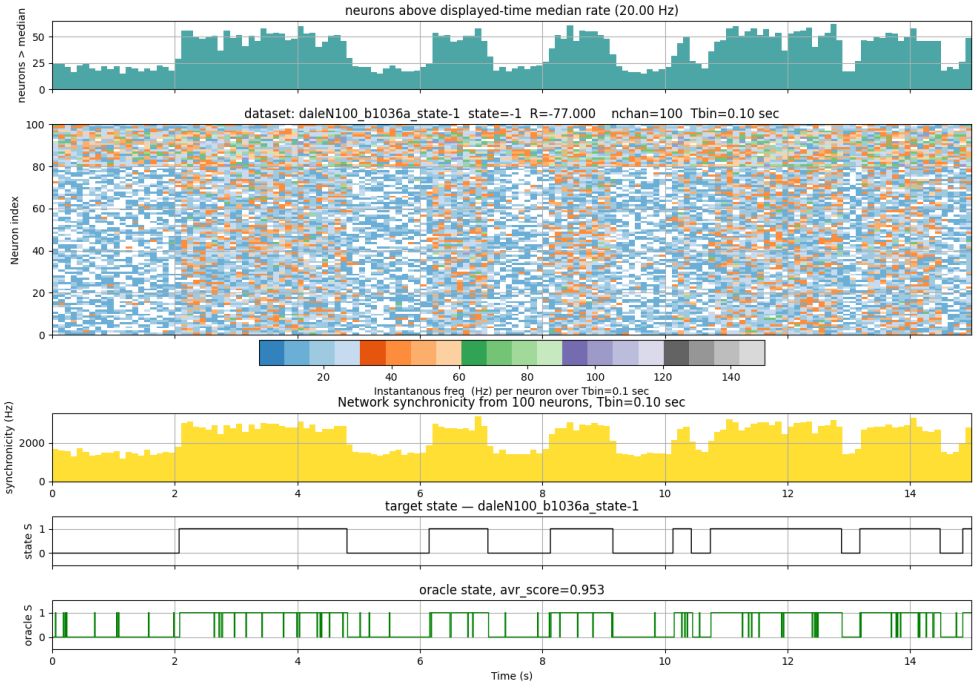

Figure 3: Ground-truth raster and latent schedule used for training/evaluation.

11.2 Training and Convergence Dynamics

Training converges quickly in EM NLL and remains spectrally stable under the $\rho(A) < 0.92$ constraint.

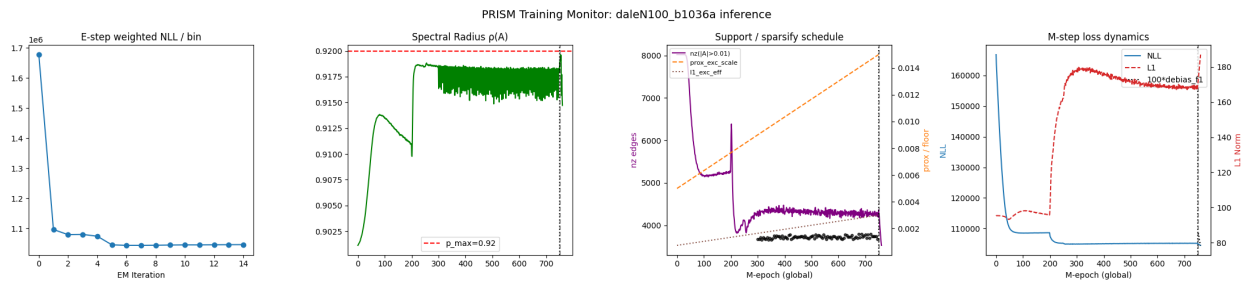

Figure 4: Training monitor. Left to right: E-step weighted NLL per EM iteration, spectral radius trajectory, support/sparsity schedule, and M-step objective dynamics.

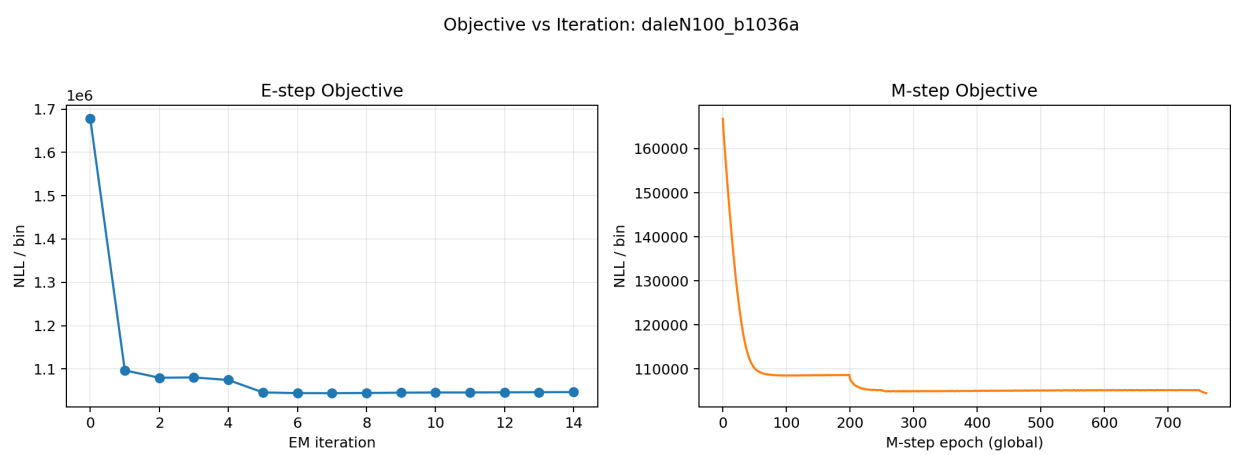

Figure 5: Objective-by-step summary. Left: EM-level E-step objective. Right: M-step objective over global epochs.

11.3 State Recovery

State recovery is high and stable, with aligned labels and near-perfect confidence.

11.4 Connectivity Recovery and Parameter Validation

Connectivity recovery is strong overall. For the run shown, the connectivity correlation is $R = 0.906$. Confusion statistics are:

$$\text{TP} = 1115, \quad \text{FP} = 126, \quad \text{FN} = 261, \quad \text{precision} = 0.898, \quad \text{recall} = 0.810, \quad F_1 = 0.852.$$

Bias recovery remains strong in both states, with $R = 0.995$ (state 0) and $R = 0.981$ (state 1).

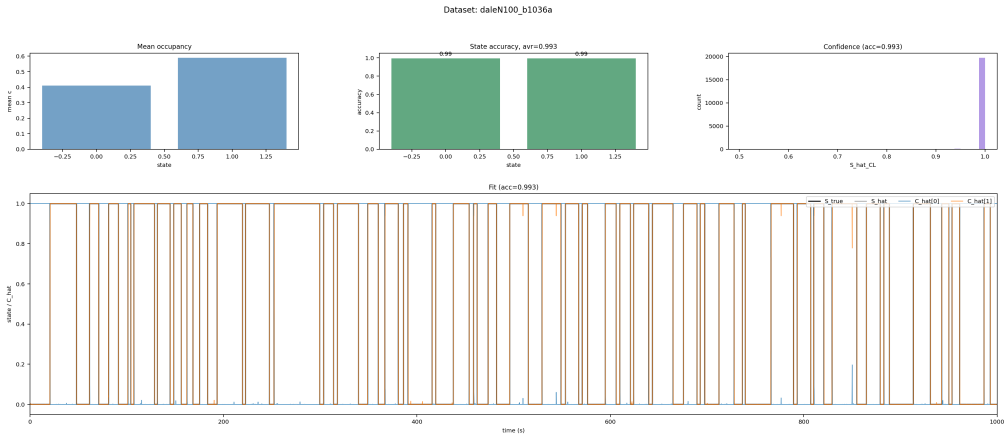

Figure 6: State recovery over time. Top row: mean occupancy, per-state accuracy, and confidence histogram. Bottom: fitted vs true state traces over the plotted time window.

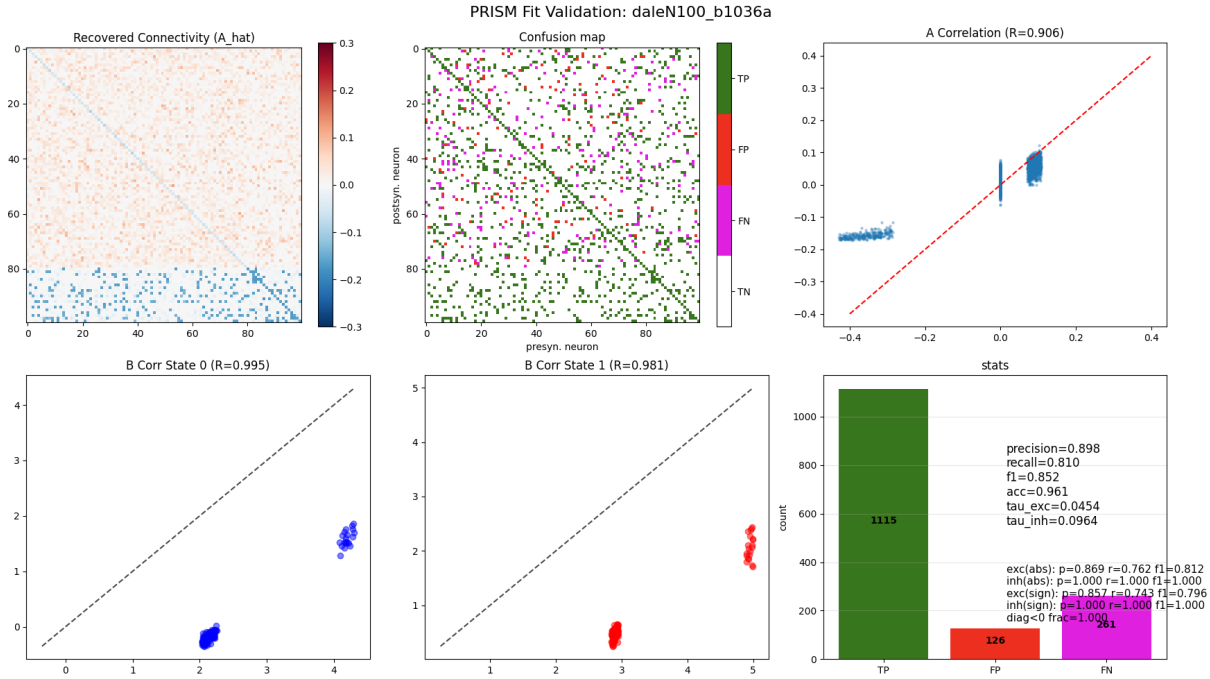

Figure 7: Connectivity validation. Top: recovered A, confusion map, and A correlation. Bottom: bias recovery scatter for state 0 and state 1 plus summary metrics.

12 Hyperparameters: Why They Exist and How Used

Name	Why it exists	Sweep/fixed and current value	Where used in math
λ_1 (`l1`)	Base sparsity strength. Higher means stronger edge shrinkage.	sweep: $\{10^{-3}, 3\cdot 10^{-3}, 10^{-2}, 3\cdot 10^{-2}, 10^{-1}, 3\cdot 10^{-1}, 1\}$	Forms $\lambda_{\text{exc}}, \lambda_{\text{inh}}$ and therefore $\tau_{\text{exc}}, \tau_{\text{inh}}$.
λ_2 (`l2`)	Controls weight magnitude smoothness and numerical stability.	sweep: $\{10^{-4}, 3\cdot 10^{-4}, 10^{-3}, 3\cdot 10^{-3}, 10^{-2}\}$	$\frac{\lambda_2}{2}\lVert A\rVert_F^2$ in Stage A and Stage B.
`l1_exc`=1.25 `l1`	Extra sparsity pressure for excitatory rows.	fixed multiplier 1.25	Base in $\lambda_{\text{exc}}^{\text{eff}}(s)$.
`l1_inh`=1.0 `l1`	Sparsity pressure for inhibitory rows.	fixed multiplier 1.0	Base in τ_{inh}.
`l1_exc_mult_start,end,power`	Ramp excitatory shrinkage over training.	fixed $(0.2, 0.8, 1.0)$	$m_{\text{exc}}(s) = m_0 + (m_1 - m_0)s^{p_m}$.
`prox_scale_exc_start,end,power`	Time varying scale for excit proximal threshold.	fixed $(5\cdot 10^{-3}, 1.5\cdot 10^{-2}, 1.0)$	$\alpha_{\text{exc}}(s)$ in τ_{exc}.
`prox_scale_inh`	Fixed proximal scale for inhibitory rows.	fixed $5\cdot 10^{-3}$	$\tau_{\text{inh}} = \lambda_{\text{inh}}\alpha_{\text{inh}}$.
`hardzero_start_frac`	Delays hard-zero so model can discover edges first.	fixed 0.40	Defines g_{start}.
`hardzero_every`	Avoids hard-zero every epoch (too harsh).	fixed 5	Periodic trigger $(g - g_{\text{start}}) \bmod q_{\text{hz}} = 0$.
`zero_floor_exc_quantile`	Makes excit hard-zero adaptive to current weight scale.	fixed 0.10	Quantile in θ_{exc}.
`zero_floor_exc_min`	Lower bound to avoid tiny/degenerate floor.	fixed 0.0	$\theta_{\text{exc}} = \max(\cdot, \theta_{\text{exc,min}})$.
`zero_floor_inh`	Optional hard-zero floor for inhibitory rows.	fixed 0.0	θ_{inh}.
`diag_inhibitory_floor`	Forces self-coupling inhibitory and not too weak.	fixed $5\cdot 10^{-2}$	$A_{ii} \leftarrow -\max(\lvert A_{ii}\rvert, d_{\min})$.
κ	Sticky states (favors staying in same state).	fixed 1.0	Added to transition diagonal in $\tilde{P}_{ij}$.
`P_pseudocount`	Prevents zero transition probabilities.	fixed 10^{-3}	ϵ_P in $\tilde{P}_{ij}$.
`pi_pseudocount`	Prevents zero initial-state probability.	fixed 10^{-2}	α_π in π update.
`gamma_temp_start,end`	Keeps E-step soft early, sharper late.	fixed $(1.6, 1.0)$	Temperature in $\log\gamma/\tau$.
`gamma_floor`	Anti-collapse for latent states.	fixed 0.02	Floor in $\gamma \leftarrow (1 - Kf)\gamma + f$.
`warm_ridge`	Stabilizes warm-start linear solve.	fixed 10^{-2}	Ridge in A_{ls}.
`warm_keep_ratio`	Controls how sparse warm-start support is.	fixed 0.15	Quantile keep rule in A_{sparse}.
`delta_t`	Converts rates to bin counts.	fixed 0.1	In $\mu = \exp(\tilde{\eta})\Delta t$ and $\log\Delta t$ term.
`debias_epochs`	Max Stage B length.	fixed 40	Outer loop count in Stage B.
`debias_lr`	Step size in Stage B.	fixed 10^{-3}	Adam learning rate for debias update.
`debias_support_eps`	Support-freeze strictness. Higher means fewer active debias edges.	fixed 10^{-3}	$S_{ij} = \mathbf{1}(\lvert A_{ij}\rvert > \varepsilon_{\text{sup}})$.
`debias_refresh_every`	Refreshes γ^{db} to reduce stale state posteriors.	fixed 5	E-step refresh period in Stage B.

Name	Why it exists	Sweep/fixed and current value	Where used in math
`debias_patience`	Stops Stage B if F1 no longer improves.	fixed 8	Early-stop rule.
`debias_min_delta`	Improvement threshold to count as real progress.	fixed 10^{-4}	$F1_{\mathrm{db}}(e) > F1_{\mathrm{best}} + \delta$.
EM iterations	Number of EM rounds.	fixed 15	E/M alternation count.
M-step epochs / EM	Stage A optimization depth per EM round.	fixed 50	Inner optimizer epochs.
batch size	Throughput vs gradient noise tradeoff.	fixed 2048	Minibatch in Stage A/B.
Stage A learning rate	Main optimizer step size.	fixed $5 \cdot 10^{-3}$	Adam in Stage A.

13 Important Current Script Behavior

The current `main()` contains explicit loop breaks after the first successful inner and outer iterations. So, even though sweep sets are defined, one run currently executes effectively one config unless those breaks are removed.